

Supercharge **K8s** Troubleshooting with AI & K8sGPT



Kosseila Hd
CloudThrill



Hi, I'm Kosseila

Cloud Dude



Cloudthrill.ca



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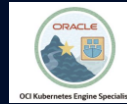
Linkedin.com/in/kousshd



GitHub.com/brokedba

- I live in Toronto 🇧🇪🇨🇦
- ~18 years infra (Oracle ACE pro)
- Blogger/Speaker
- IaC/DevOps enthusiast
- Founder, @CloudThrill
- Podcaster, Beatmaker 🎧🎵🎧

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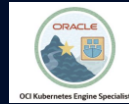


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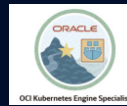
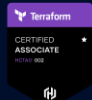


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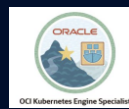
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Tech Beats



Kosseila Hd Principal Consultant @CloudThrill



AGENDA

1- Challenges (context)

- ❑ AI vs traditional troubleshooting methods
- ❑ Common challenges in managing in Kubernetes clusters

2- Introduction to K8sGPT

- ❑ What is K8sGPT?
- ❑ Key features & benefits
- ❑ Specs and installation for K8s

3- Run K8sGPT with OKE

- ❑ K8sGPT commands
- ❑ K8sGPT Operator architecture
- ❑ Demo in in OKE & What's Next

4- Q&A



Engineers*



**MARK ZUCKERBERG SAYS THIS
YEAR META'S AI WILL CODE LIKE
ENGINEERS, REPLACING HUMANS
AND BUILDING MOST APP CODE**



Engine



\$50 Billions MetaFlop

**ONLY WAY TO KEEP
AI FROM TAKING YOUR JOB**

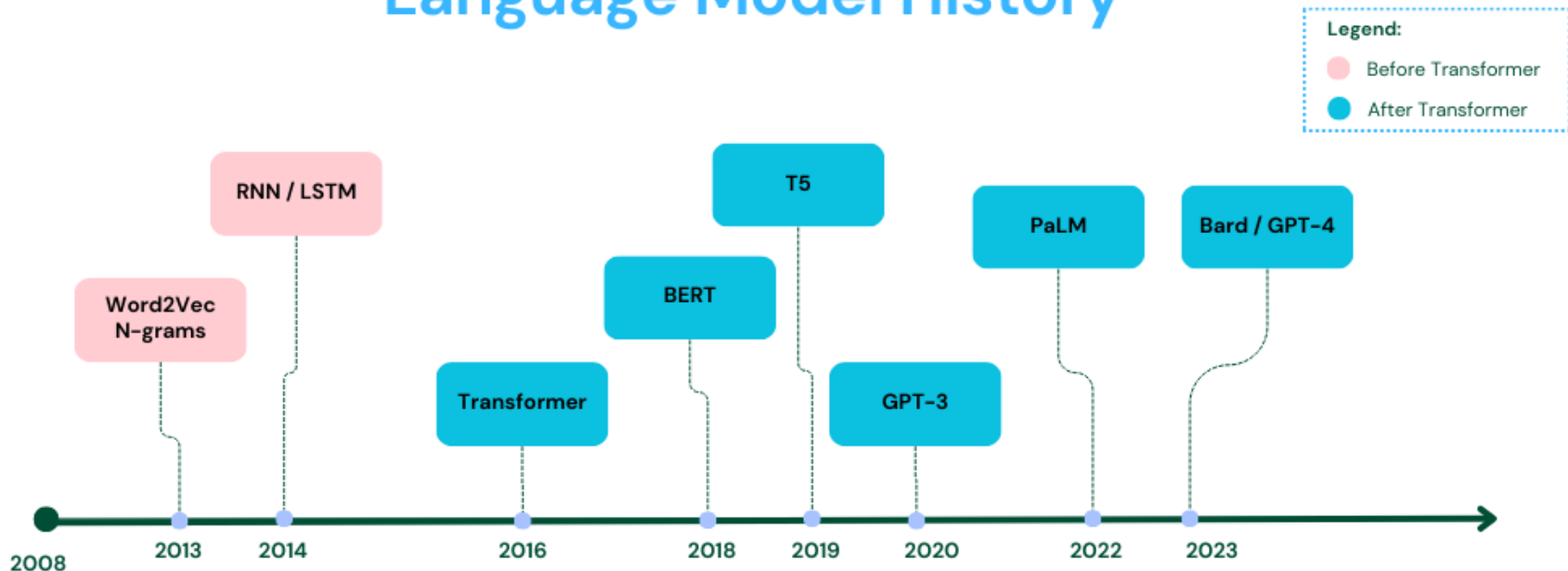
**IS TO USE AI TO
DO YOUR JOB BETTER**



GPT Dependence ⚠

Tectonic shift in AI – (Transformers) timeline

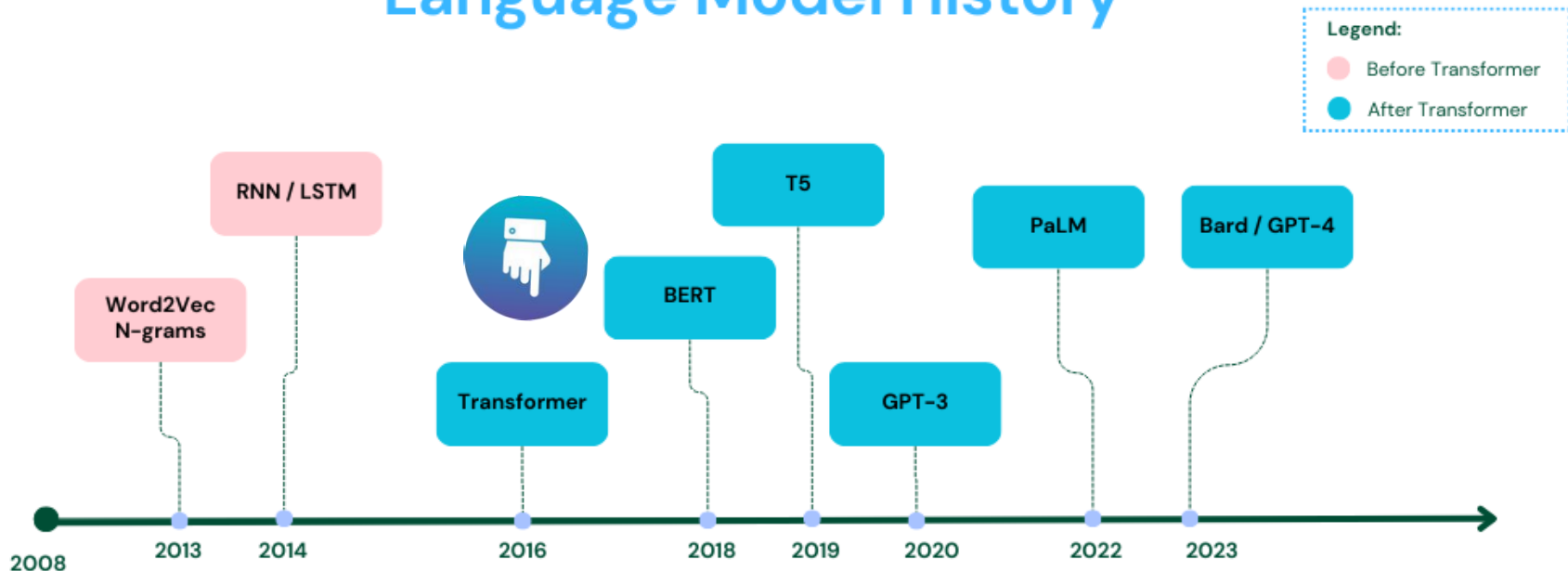
Language Model History



[LLM History article](#) 

Tectonic shift in AI – (Transformers) timeline

Language Model History



[LLM History article](#) 

The Challenges

What AI Solves

- Pattern recognition
- Translation
- Prediction

The Challenges

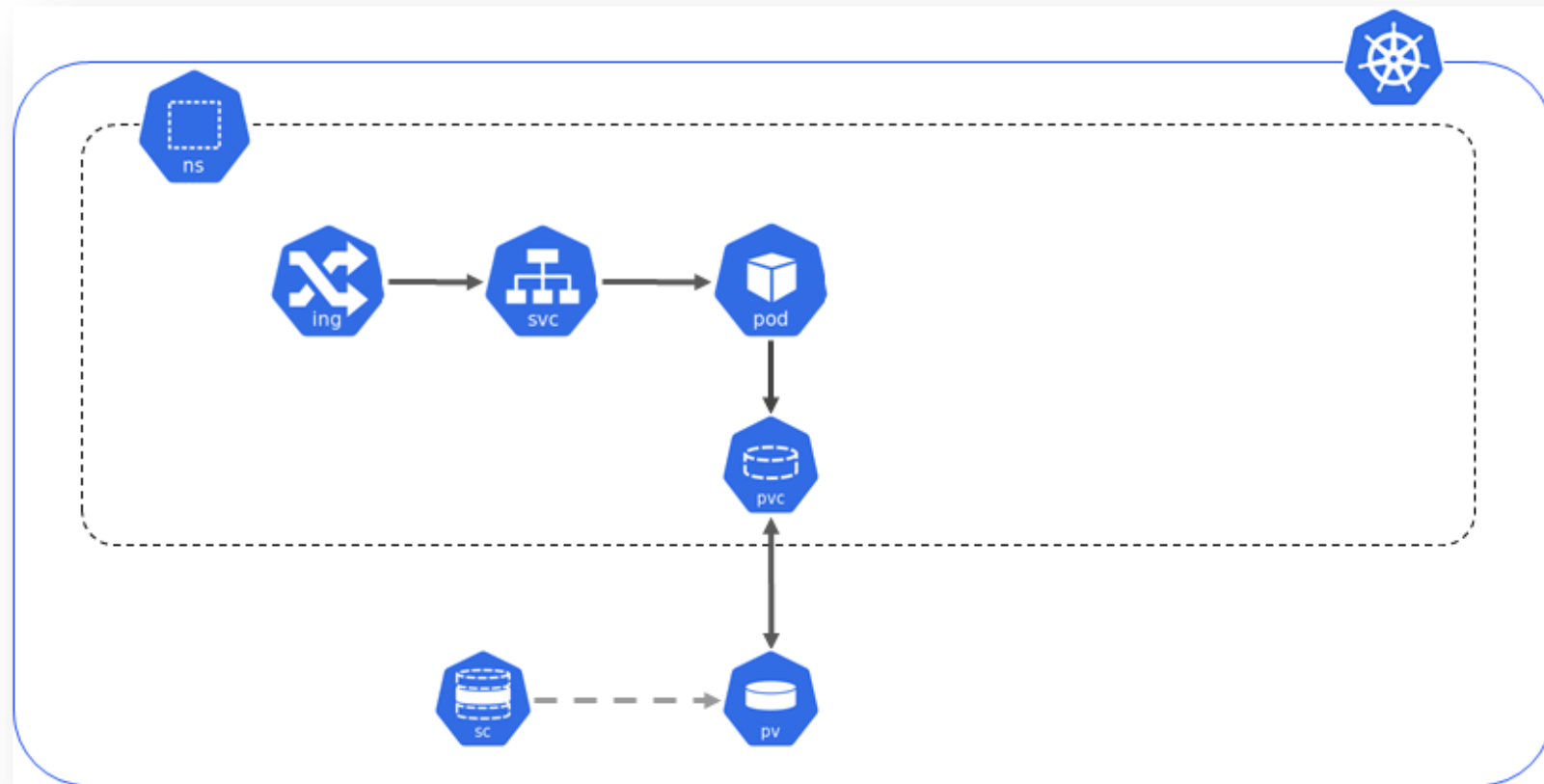
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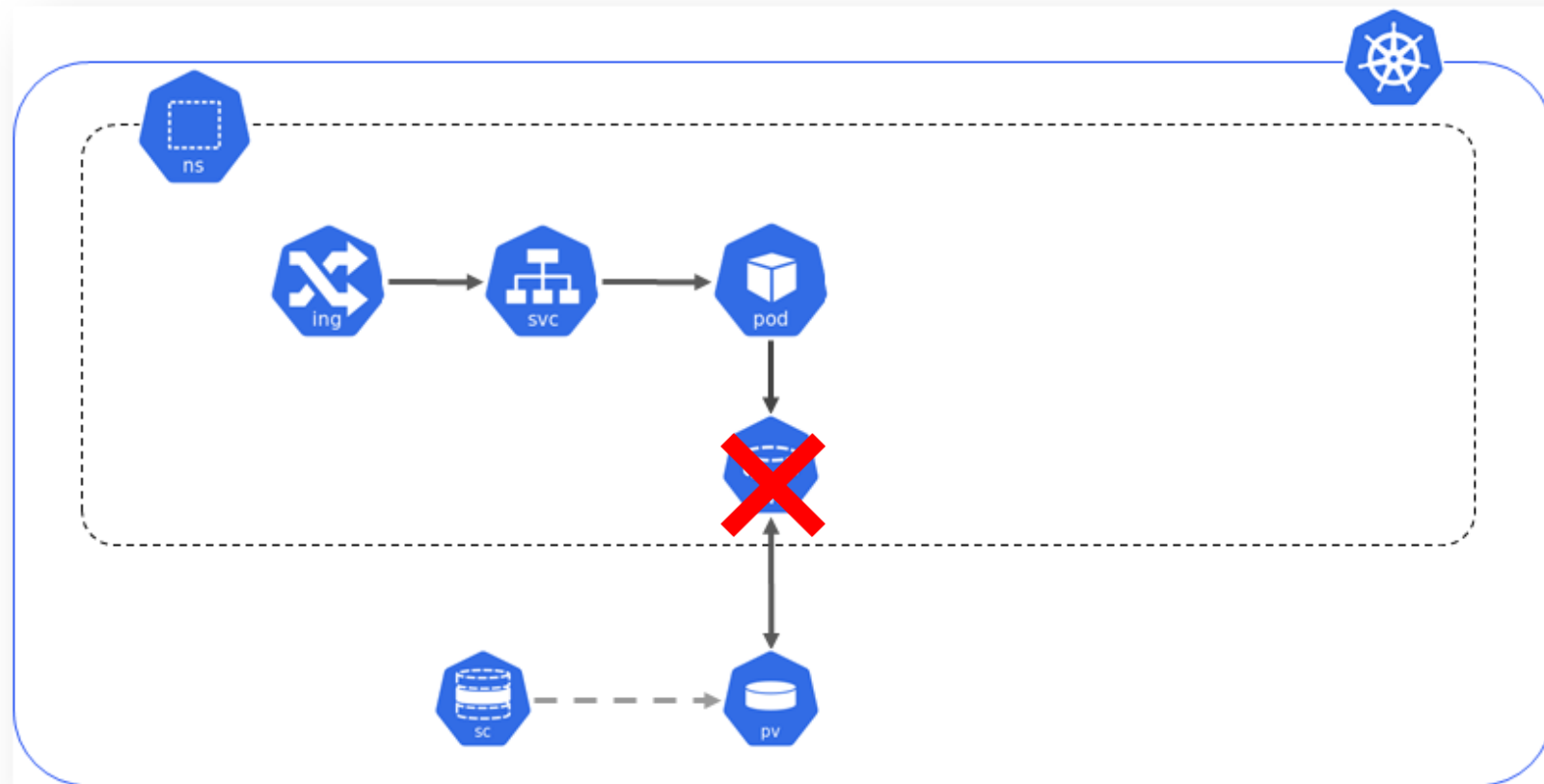
K8s struggles

- Workload/node failure
- Error Correlation across stack
- Forecasting /capacity planning

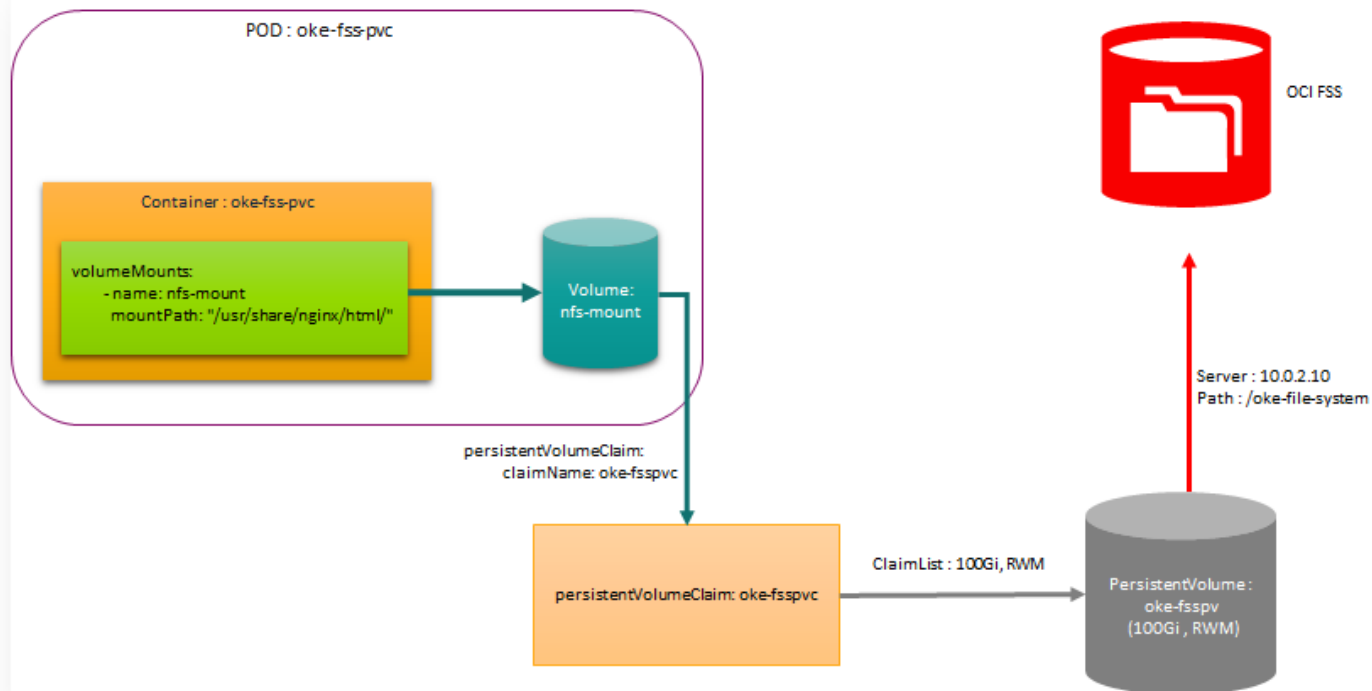
The Challenges



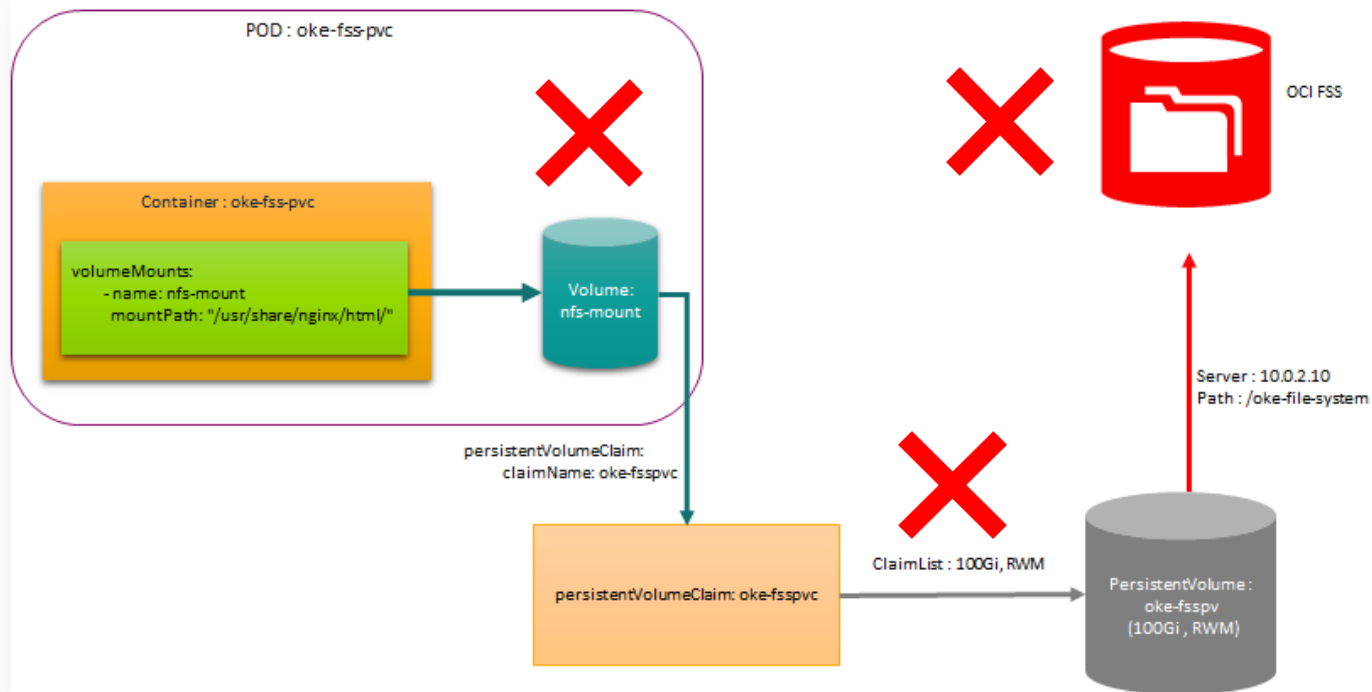
The Challenges



The Challenges



The Challenges



The Challenges

- Troubleshooting K8 is hard
- Error messages are complex
- Signal/Noise ratio
- Remediation takes time
- N-dimensional layers
- Knowledge retention in SRE's head



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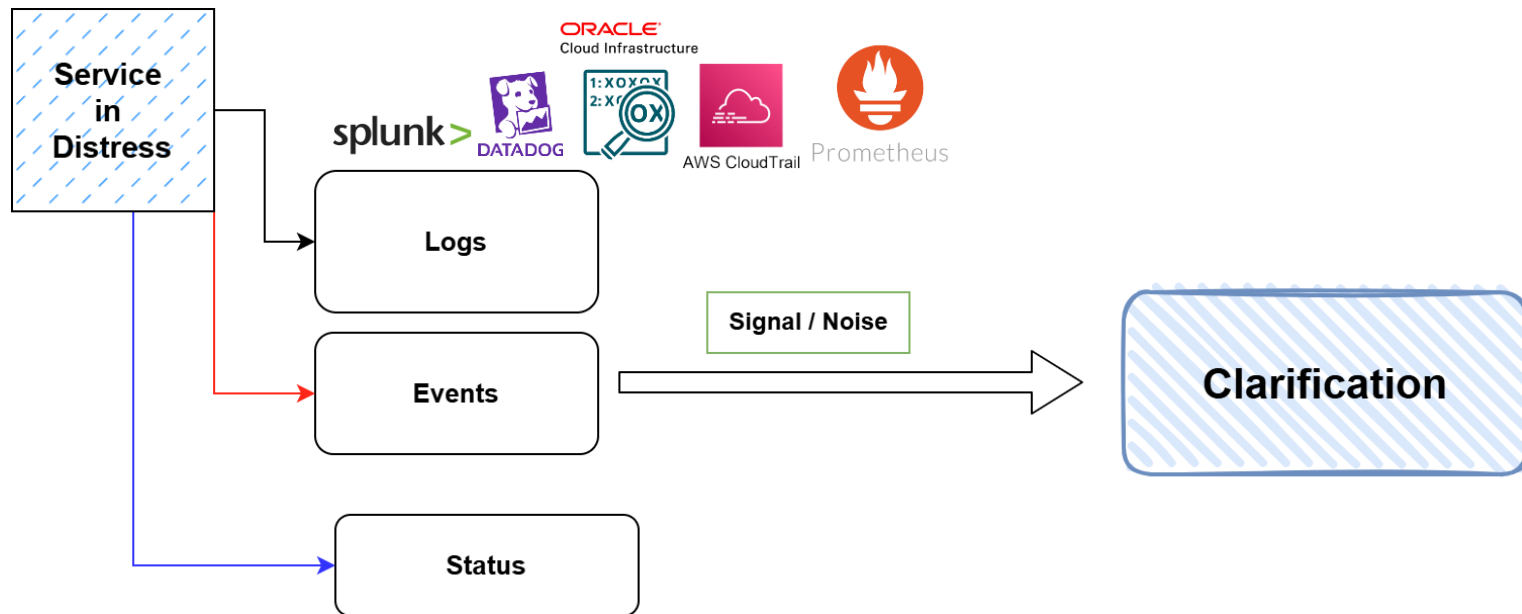


[● ◀] systemd

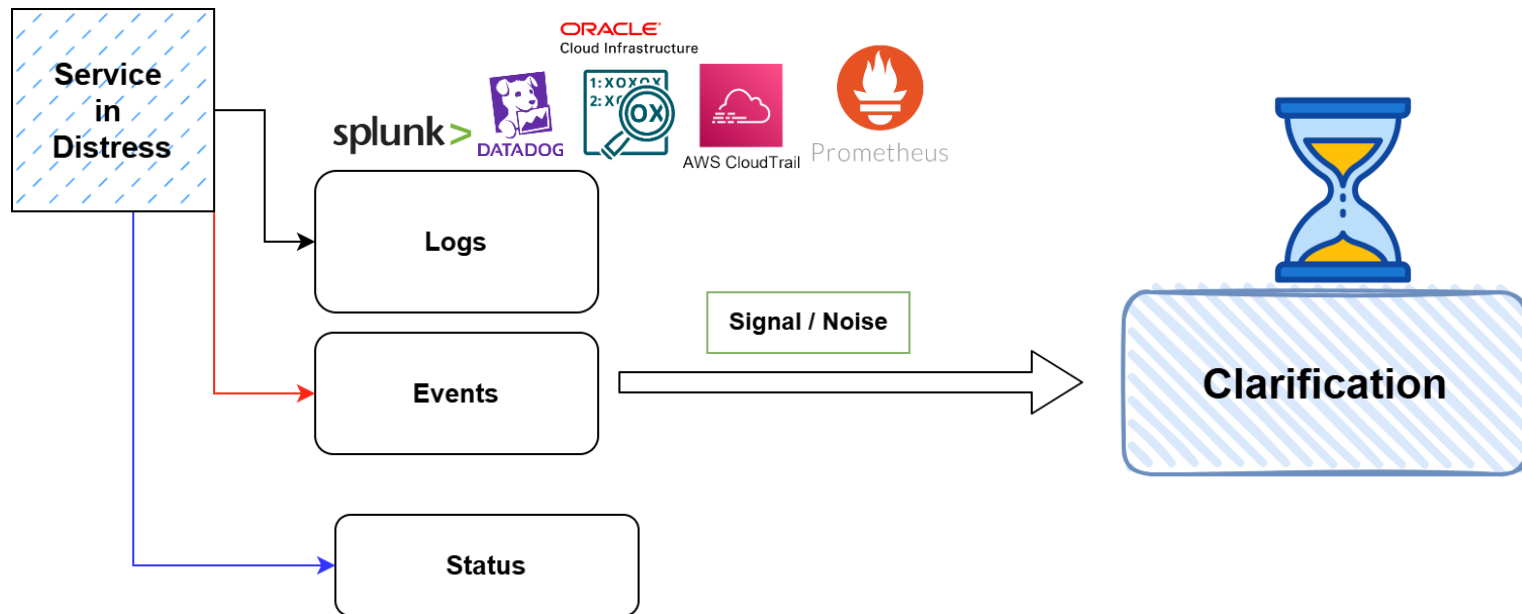
● Syslog



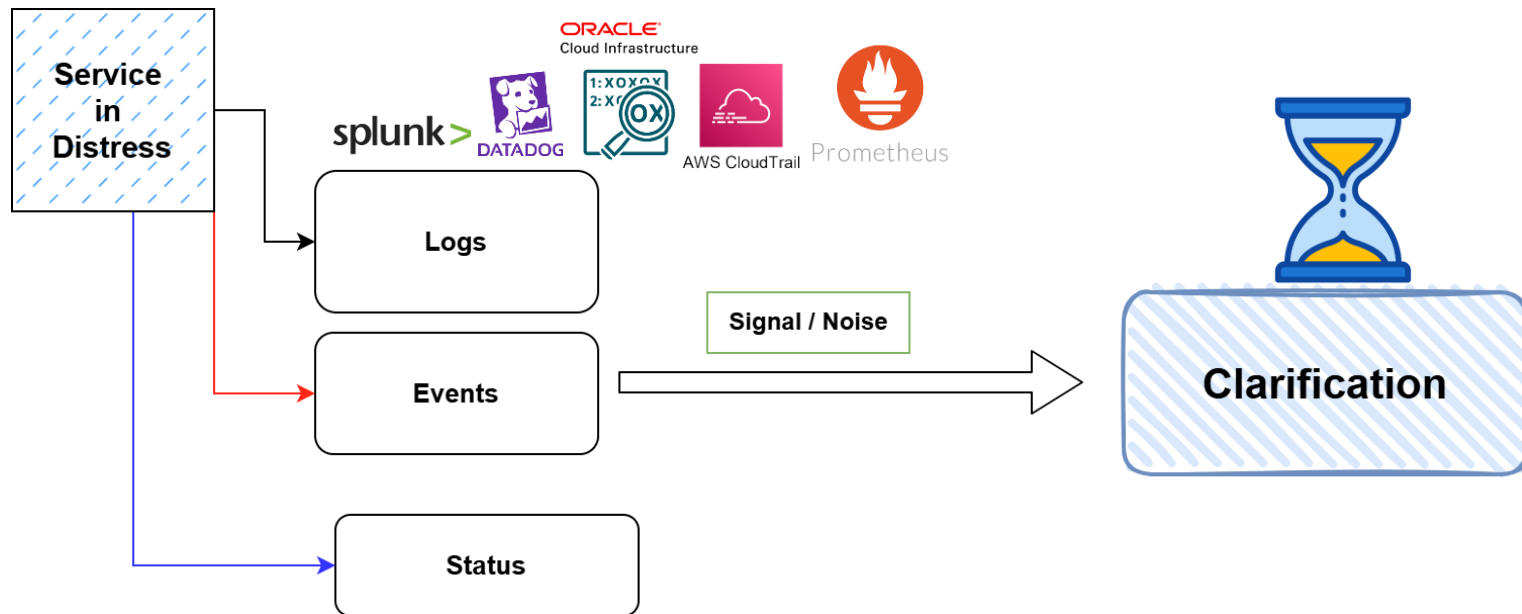
The Challenges



The Challenges



The Challenges



What can we do about it?

Introducing K8sGPT

What is

Tool for scanning your Kubernetes clusters, diagnosing, and triaging issues in simple English.



Introducing K8sGPT

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Tool for scanning your Kubernetes clusters, diagnosing, and triaging issues in simple English.

In a nutshell:

1. OpenSource, 92 contributors, (6k Git Stars)
2. Fast triage, AI analysis
3. SRE experience codified into its 19 analyzers
4. Convert Complex signals into fixes
5. Integrates 12 AI sources (OpenAI, Cohere, ollama...)
6. Integration with Security CVE scanners (**Trivy**)



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K8sGPT gives Kubernetes Superpowers to everyone!

Key features

Work smart, not hard

- Codified SRE knowledge knows what to look for
- **AI** cuts through the noise **but that's not all!**
- Can run from a **CLI** or **Operator**

```
k8sgpt on main via v1.20.2  
> |
```

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Analyzers {Filters}

Anonymization

Remote caching

Integrations(trivy, Prometheus)

LLM AI Backends

Key features Work smart, not hard

19

```
brokedba@Orpheus:~$ k8sgpt filters list
```

Active:

- > ReplicaSet
- > Service
- > Ingress
- > StatefulSet
- > CronJob
- > Node
- > ValidatingWebhookConfiguration
- > Pod
- > MutatingWebhookConfiguration
- > PersistentVolumeClaim
- > Deployment

Unused:

- > PodDisruptionBudget
- > NetworkPolicy
- > Log
- > GatewayClass
- > Gateway
- > HTTPRoute
- > HorizontalPodAutoScaler



Analyzers {Filters}

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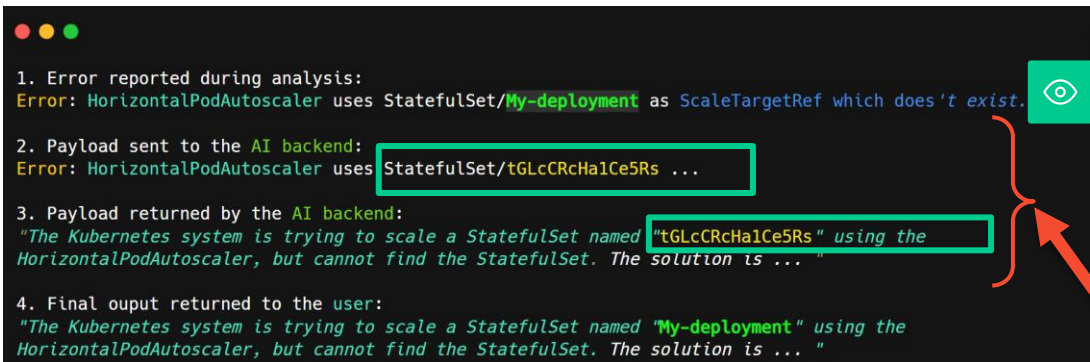
Integrations(trivy, Prometheus)

LLM AI Backends

** Custom analyzers are supported*

Key features Work smart, not hard

☐ Original data (unmasked)



A terminal window with a dark background and light green text. It displays four numbered steps of an analysis process. Step 1 shows an error about a missing ScaleTargetRef. Step 2 shows a payload sent to an AI backend, with a red box highlighting the StatefulSet name. Step 3 shows the payload returned by the AI backend, with a red box highlighting the StatefulSet name. Step 4 shows the final output returned to the user, with a red box highlighting the StatefulSet name. A red eye icon is located to the right of the terminal window, with a red arrow pointing from it to the red boxes in the terminal output.

```
1. Error reported during analysis:
Error: HorizontalPodAutoscaler uses StatefulSet/My-deployment as ScaleTargetRef which doesn't exist.

2. Payload sent to the AI backend:
Error: HorizontalPodAutoscaler uses StatefulSet/tGLcCRcHa1Ce5Rs ...

3. Payload returned by the AI backend:
"The Kubernetes system is trying to scale a StatefulSet named "tGLcCRcHa1Ce5Rs" using the
HorizontalPodAutoscaler, but cannot find the StatefulSet. The solution is ..."

4. Final output returned to the user:
"The Kubernetes system is trying to scale a StatefulSet named "My-deployment" using the
HorizontalPodAutoscaler, but cannot find the StatefulSet. The solution is ..."
```

Analzers {Filters}



Anonymization

Remote caching

Integrations(trivy, Prometheus)

LLM AI Backends

☐ Masked data sent to AI

Key features Work smart, not hard

☐ Original data (unmasked)

Analyzers {Filters}



Anonymization

Remote caching

Integrations(trivy, Prometheus)

LLM AI Backends

☐ Masked data sent to AI

☐ Solution returned to the user with unmasked data

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Key features Work smart, not hard

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Analyzers {Filters}



Anonymization

Remote caching

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LLM AI Backends

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```

Some analyzers are still not masked (events)

Key features

Work smart, not hard

- ❖ Remote caching offloads cached data to a remote storage service for consistency and to reduce local storage

Analyzers {Filters}

Anonymization



Remote caching

Integrations(trivy, Prometheus)

LLM AI Backends

Key features

Work smart, not hard

- **Prometheus Integration:** K8sGPT leverages real-time metrics from Prometheus.
- **Trivy Integration:** leverages CVE analysis from trivy's continuous scanning for vulnerabilities.



Analyzers {Filters}

Anonymization

Remote caching



Integrations(trivy, Prometheus)

LLM AI Backends

Key features Work smart, not hard

- **Prometheus Integration:** K8sGPT leverages real-time metrics from Prometheus. 
- **Trivy Integration:** leverages CVE analysis from trivy's continuous scanning for vulnerabilities. 

```
k8sgpt integration activate trivy

> k8sgpt filters list

Active:
> VulnerabilityReport (integration)
> Pod
> ConfigAuditReport (integration)
Unused:

> k8sgpt analyze --filter VulnerabilityReport
```

Analyzers {**Filters**}

Anonymization

Remote caching



Integrations(trivy, Prometheus)

LLM AI Backends

Key features Work smart, not hard

```
brokedba@Orpheus:~$ k8sgpt auth list
```

Default:

> openai

Active:

Unused:

> openai

> localai

> ollama

> azureopenai

> cohere

> amazonbedrock

> amazonsagemaker

> google

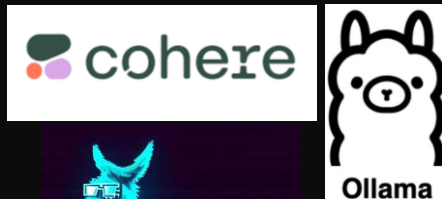
> noopai

> huggingface

> googlevertexai

> oci

> watsonxai



Analyzers {**Filters**}

Anonymization

Remote caching

Integrations(trivy, Prometheus)



LLM AI Backends

Key features Work smart, not hard



Analyzers {**Filters**}

Anonymization

Remote caching

Integrations(trivy, Prometheus)



LLM AI Backends

```
1. Setup OCI CLI config under .oci/  
2. Add Backend  
k8sgpt auth add --backend oci --model <Model OCID> --compartmentId <Compartment OCID>  
3. Run a test  
k8sgpt analyze --explain --backend oci
```

K8sGPT: CLI Installation

❑ RHEL/ OracleLinux

```
sudo rpm -ivh k8sgpt\_amd64.rpm
```

❑ Mac

```
brew install k8sgpt
```

❑ Ubuntu

```
sudo dpkg -i k8sgpt_amd64.deb
```

Windows (WSL use apt-get)

```
 k8sgpt\_Windows\_x86\_64.zip
```

K8sGPT: CLI Installation



P Personal ▾ / K8sGPT

 DASHBOARD

 Assistants

 Fine-tuning

 Batches

 Storage

 Usage

 API keys

P Personal ▾ / K8sGPT

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 Usage

 **API keys**

Create new secret key

Owned by

☒ You ☐ Service account

This API key is tied to your user and can make requests against the selected project. If you are removed from the organization or project, this key will be disabled.

Name Optional

k8sgpt_key

Project

K8sGPT ▾

Permissions

All Restricted Read Only

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 **API keys**

API keys



Project API keys have replaced user API keys.

We recommend using project based API keys for more granular control over your resources. [Learn more](#)

NAME	SECRET KEY	CREATED	PERMISSIONS
k8sgpt_key	sk-...Ge8A	Aug 22, 2024	All

K8sGPT: CLI Installation



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✓ Add OpenAI

```
$ k8sgpt auth add -b openai
```

Warning: model input is empty, will use the default value: gpt-3.5-turbo

Enter openai Key: openai added to the AI backend provider list

K8sGPT: CLI Installation



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✓ Check authentication

```
$ k8sgpt auth list
```

Active:

```
> openai
```

K8sGPT: CLI Installation



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✅ **Analyze**

```
$ k8sgpt analyze --explain -b openai
```

K8sGPT: CLI Installation



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NAME	SECRET KEY	CREATED	PERMISSIONS
k8sgpt_key	sk- . . . Ge8A	Aug 22, 2024	All

Config file locations:

OS	Path
MacOS	~/Library/Application Support/k8sgpt/k8sgpt.yaml
Linux	~/.config/k8sgpt/k8sgpt.yaml
Windows	%LOCALAPPDATA%/k8sgpt/k8sgpt.yaml

K8sGPT: CLI Installation



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API key is stored in plain text

K8sGPT: CLI Installation



P Personal ▾ / K8sGPT

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k8sgpt_key	sk-...Ge8A	Aug 22, 2024	All

```
brokedba@Orpheus:~$ cat ~/.config/k8sgpt/k8sgpt.yaml
ai:
  providers:
    - name: openai
      model: gpt-3.5-turbo
      password: sk-oGt2
      temperature: 0.7
      topp: 0.5
      topk: 50
      maxtokens: 2048
      customheaders: []
  defaultprovider: ""
```



API key is stored in plain text

Commands

```
brokedba@$ k8sgpt help

Usage:
  k8sgpt [command]

Available Commands:
  analyze    This command will find problems within your Kubernetes cluster
  auth       Authenticate with your chosen backend
  cache      For working with the cache the results of an analysis
  completion Generate the autocompletion script for the specified shell
  filters    Manage filters for analyzing Kubernetes resources
  generate   Generate Key for your chosen backend (opens browser)
  help       Help about any command
  integration Integrate another tool into K8sGPT
  serve      Runs k8sgpt as a server
  version    Print the version number of k8sgpt

Flags:
  --config string    Default config file (/home/brokedba/.config/k8sgpt/k8sgpt.yaml)
  -h, --help         help for k8sgpt
  --kubeconfig string Path to a kubeconfig. Only required if out-of-cluster.
  --kubecore string   Kubernetes context to use. Only required if out-of-cluster.

Use "k8sgpt [command] --help" for more information about a command.
```

Commands

--auth

add, list, remove AI backend

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analyze

find problems in your cluster

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Commands

--auth

add, list, remove AI backend

analyze

find problems in your cluster

-f, --filter

Filter through analyzers list

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Use "k8sgpt [command] --help" for more information about a command.
```

Commands

--auth

add, list, remove AI backend

analyze

find problems in your cluster

-f, --filter

Filter through analyzers list

-e, --explain

Explain through AI

```
brokedba@$ k8sgpt help

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  generate   Generate Key for your chosen backend (opens browser)
  help      Help about any command
  integration Integrate another tool into K8sGPT
  serve      Runs k8sgpt as a server
  version    Print the version number of k8sgpt

Flags:
  --config string    Default config file (/home/brokedba/.config/k8sgpt/k8sgpt.yaml)
  -h, --help         help for k8sgpt
  --kubeconfig string Path to a kubeconfig. Only required if out-of-cluster.
  --kubecore string   Kubernetes context to use. Only required if out-of-cluster.

Use "k8sgpt [command] --help" for more information about a command.
```

Commands

--auth

add, list, remove AI backend

analyze

find problems in your cluster

-f, --filter

Filter through analyzers list

-e, --explain

Explain through AI

-b, --backend

Backend AI provider to use
(i.e cohere)

```
brokedba@$ k8sgpt help

Usage:
  k8sgpt [command]

Available Commands:
  analyze  This command will find problems within your Kubernetes cluster
  auth     Authenticate with your chosen backend
  cache    For working with the cache the results of an analysis
  completion  Generate the autocompletion script for the specified shell
  filters  Manage filters for analyzing Kubernetes resources
  generate  Generate Key for your chosen backend (opens browser)
  help     Help about any command
  integration  Integrate another tool into K8sGPT
  serve    Runs k8sgpt as a server
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Flags:
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  -h, --help      help for k8sgpt
  --kubeconfig string  Path to a kubeconfig. Only required if out-of-cluster.
  --kubecore string  Kubernetes context to use. Only required if out-of-cluster.

Use "k8sgpt [command] --help" for more information about a command.
```

Commands

--auth

add, list, remove AI backend

analyze

find problems in your cluster

Integration

Integrate trivy or prometheus

-e, --explain

Explain through AI

-b, --backend

Backend AI provider to use
(i.e cohere)

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brokedba@$ k8sgpt help

Usage:
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Available Commands:
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  -h, --help      help for k8sgpt
  --kubeconfig string  Path to a kubeconfig. Only required if out-of-cluster.
  --kubecore string  Kubernetes context to use. Only required if out-of-cluster.

Use "k8sgpt [command] --help" for more information about a command.
```

Commands

--auth

add, list, remove AI backend

analyze

find problems in your cluster

Integration

Integrate trivy or prometheus

-e, --explain

Explain through AI

-d, --with-doc

Show official k8 documentation

```
brokedba@$ k8sgpt help

Usage:
  k8sgpt [command]

Available Commands:
  analyze  This command will find problems within your Kubernetes cluster
  auth     Authenticate with your chosen backend
  cache    For working with the cache the results of an analysis
  completion  Generate the autocompletion script for the specified shell
  filters  Manage filters for analyzing Kubernetes resources
  generate  Generate Key for your chosen backend (opens browser)
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  integration  Integrate another tool into K8sGPT
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  version  Print the version number of k8sgpt

Flags:
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  -h, --help      help for k8sgpt
  --kubeconfig string  Path to a kubeconfig. Only required if out-of-cluster.
  --kubecore string  Kubernetes context to use. Only required if out-of-cluster.

Use "k8sgpt [command] --help" for more information about a command.
```

Commands

--auth

add, list, remove AI backend

analyze

find problems in your cluster

Integration

Integrate trivy or prometheus

-a, --anonymize

Anonymize data before sending to AI

-d, --with-doc

Show official k8 documentation

```
brokedba@$ k8sgpt help

Usage:
  k8sgpt [command]

Available Commands:
  analyze   This command will find problems within your Kubernetes cluster
  auth      Authenticate with your chosen backend
  cache     For working with the cache the results of an analysis
  completion Generate the autocompletion script for the specified shell
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  --kubeconfig string Path to a kubeconfig. Only required if out-of-cluster.
  --kubecore string   Kubernetes context to use. Only required if out-of-cluster.

Use "k8sgpt [command] --help" for more information about a command.
```

Commands

--auth

add, list, remove AI backend

-n, --namespace

Namespace to analyze

Integration

Integrate trivy or prometheus

-a, --anonymize

Anonymize data before sending to AI

-d, --with-doc

Show official k8 documentation

```
brokedba@$ k8sgpt help

Usage:
  k8sgpt [command]

Available Commands:
  analyze   This command will find problems within your Kubernetes cluster
  auth      Authenticate with your chosen backend
  cache     For working with the cache the results of an analysis
  completion Generate the autocompletion script for the specified shell
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  --config string    Default config file (/home/brokedba/.config/k8sgpt/k8sgpt.yaml)
  -h, --help         help for k8sgpt
  --kubeconfig string Path to a kubeconfig. Only required if out-of-cluster.
  --kubecore string  Kubernetes context to use. Only required if out-of-cluster.

Use "k8sgpt [command] --help" for more information about a command.
```


Commands

--auth

add, list, remove AI backend

-n, --namespace

Namespace to analyze

Integration

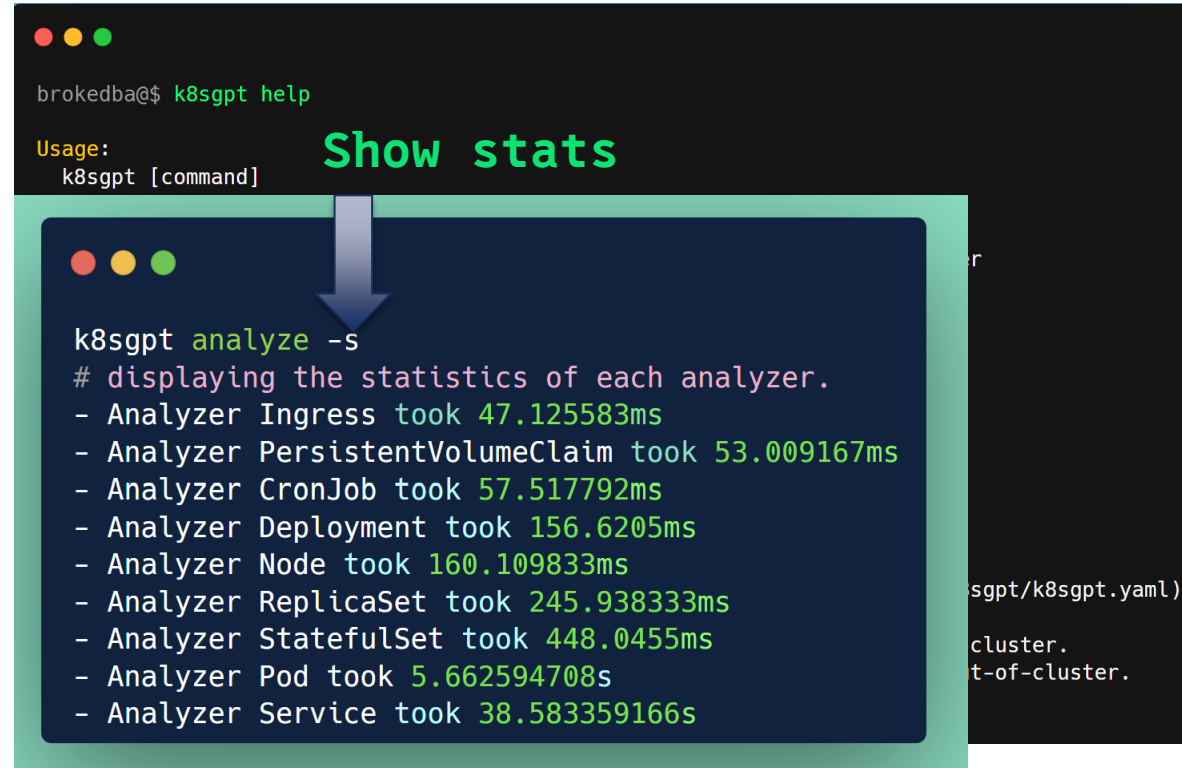
Integrate trivy or prometheus

-a, --anonymize

Anonymize data before sending to AI

-d, --with-doc

Show official k8 documentation



```
brokedba@$ k8sgpt help

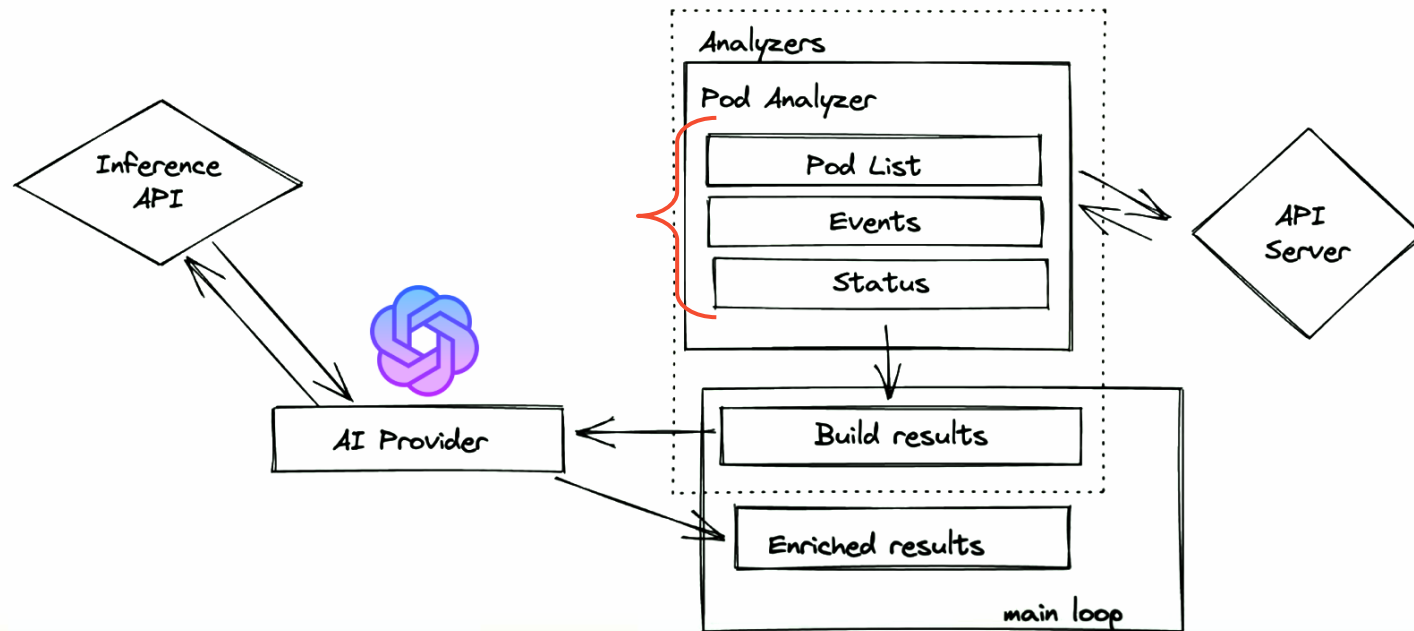
Usage:
k8sgpt [command]

Show stats

k8sgpt analyze -s
# displaying the statistics of each analyzer.
- Analyzer Ingress took 47.125583ms
- Analyzer PersistentVolumeClaim took 53.009167ms
- Analyzer CronJob took 57.517792ms
- Analyzer Deployment took 156.6205ms
- Analyzer Node took 160.109833ms
- Analyzer ReplicaSet took 245.938333ms
- Analyzer StatefulSet took 448.0455ms
- Analyzer Pod took 5.662594708s
- Analyzer Service took 38.583359166s
```

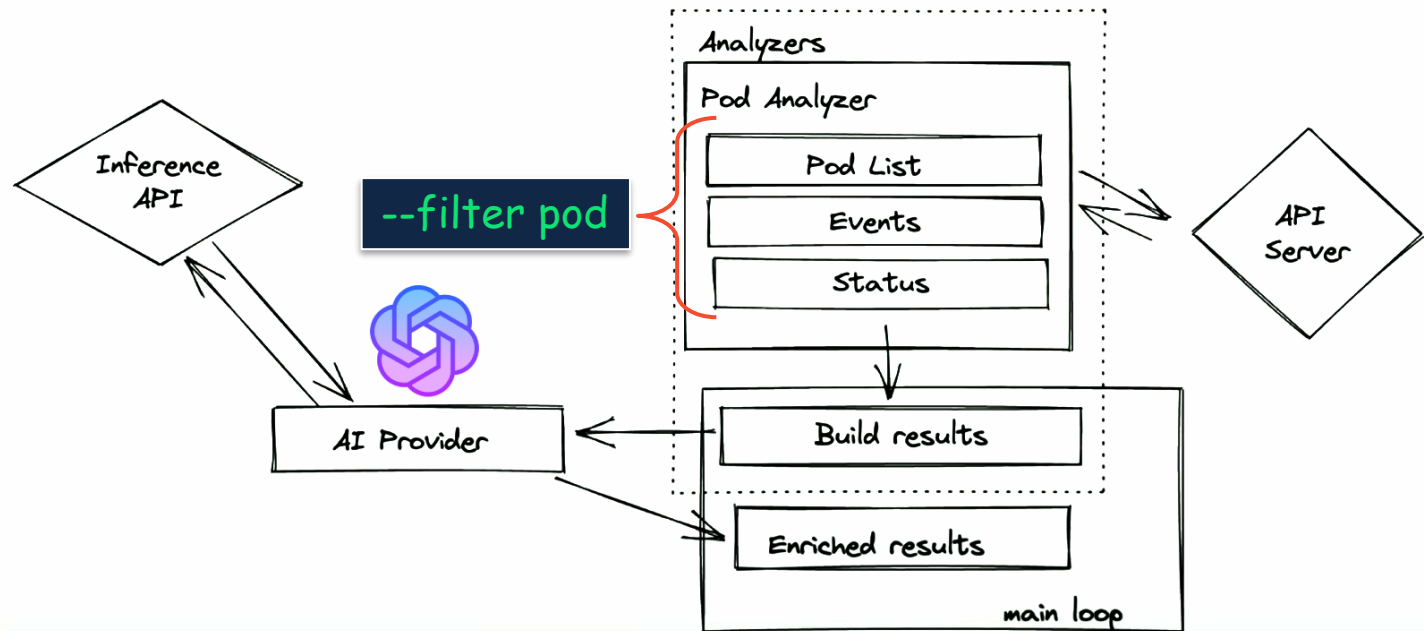
K8SGPT (Operator): Architecture

□ Basic K8SGPT (CLI)



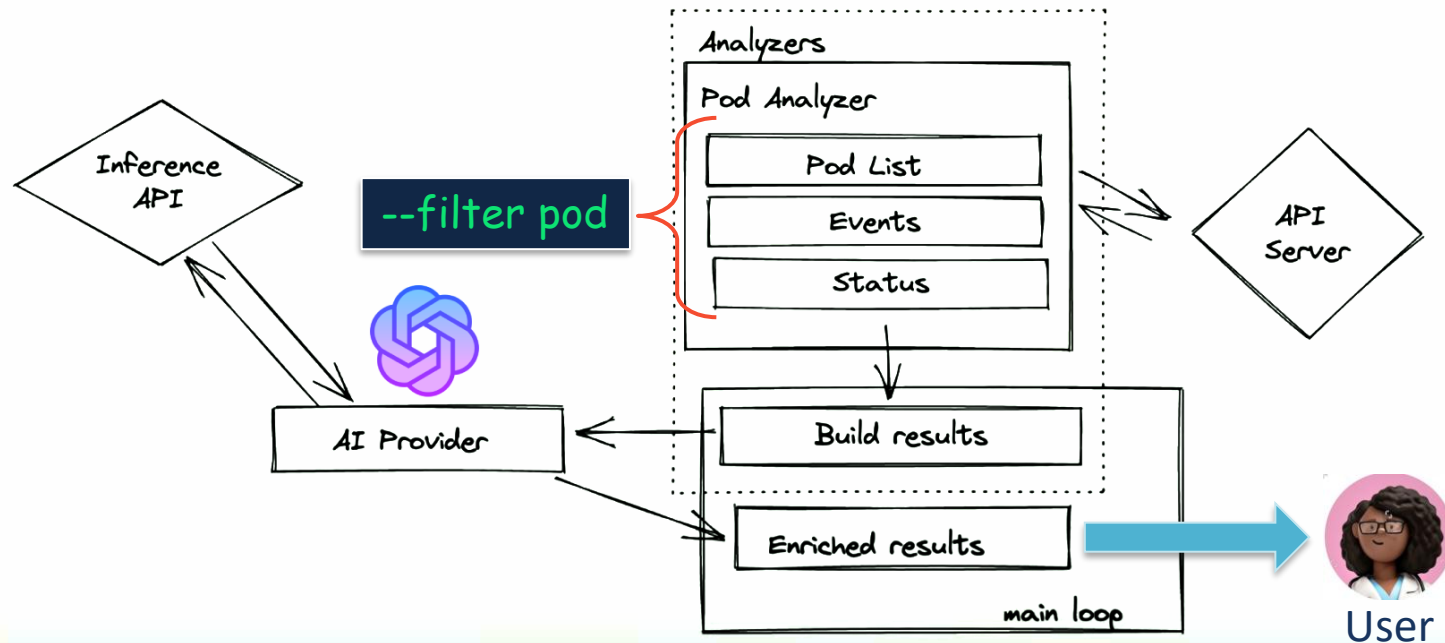
K8SGPT (Operator): Architecture

❑ Basic K8SGPT (CLI)



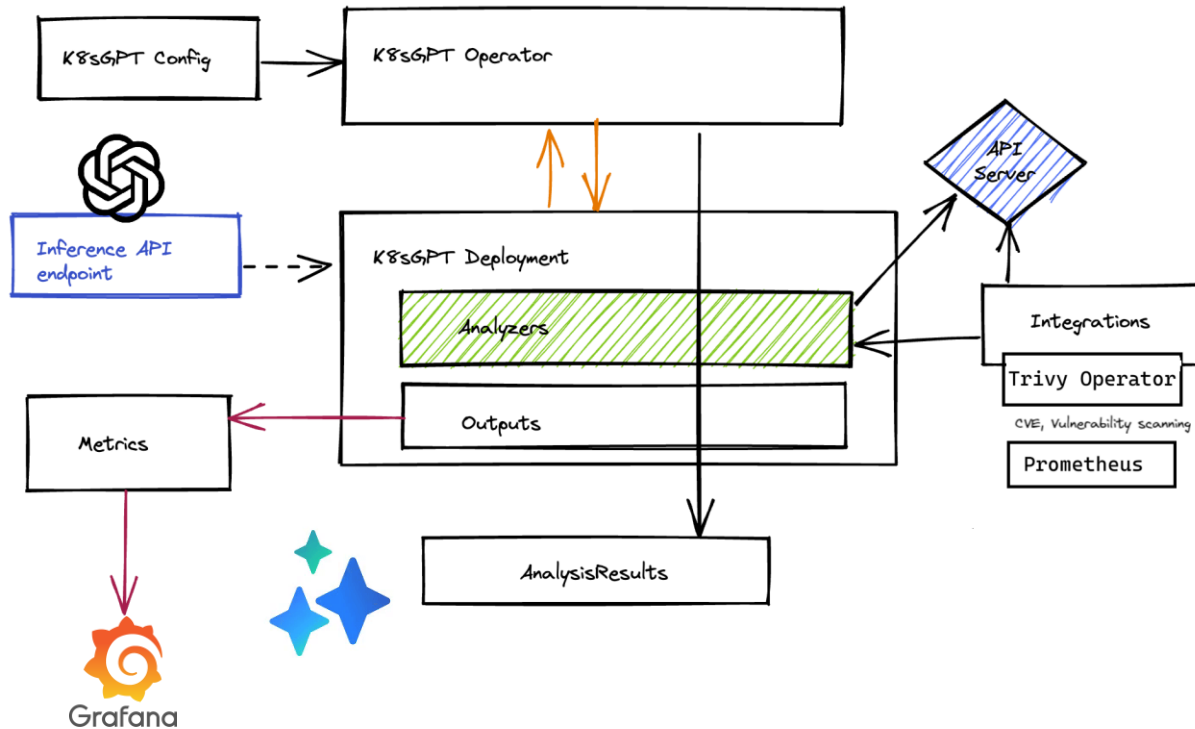
K8SGPT (Operator): Architecture

□ Basic K8SGPT (CLI)



K8SGPT (Operator): Architecture

❑ K8SGPT Operator architecture



K8SGPT (Operator): Architecture

```
Editor  Tab 1  Tab 2  +
Context: kubernetes-admin@kubernetes
Cluster: kubernetes
User: kubernetes-admin
K9s Rev: v0.32.5 < v0.32.7
K8s Rev: v1.31.0
CPU: 12%
MEM: 73%

Describe(k8sgpt-operator-system/defaultbrokenpod)

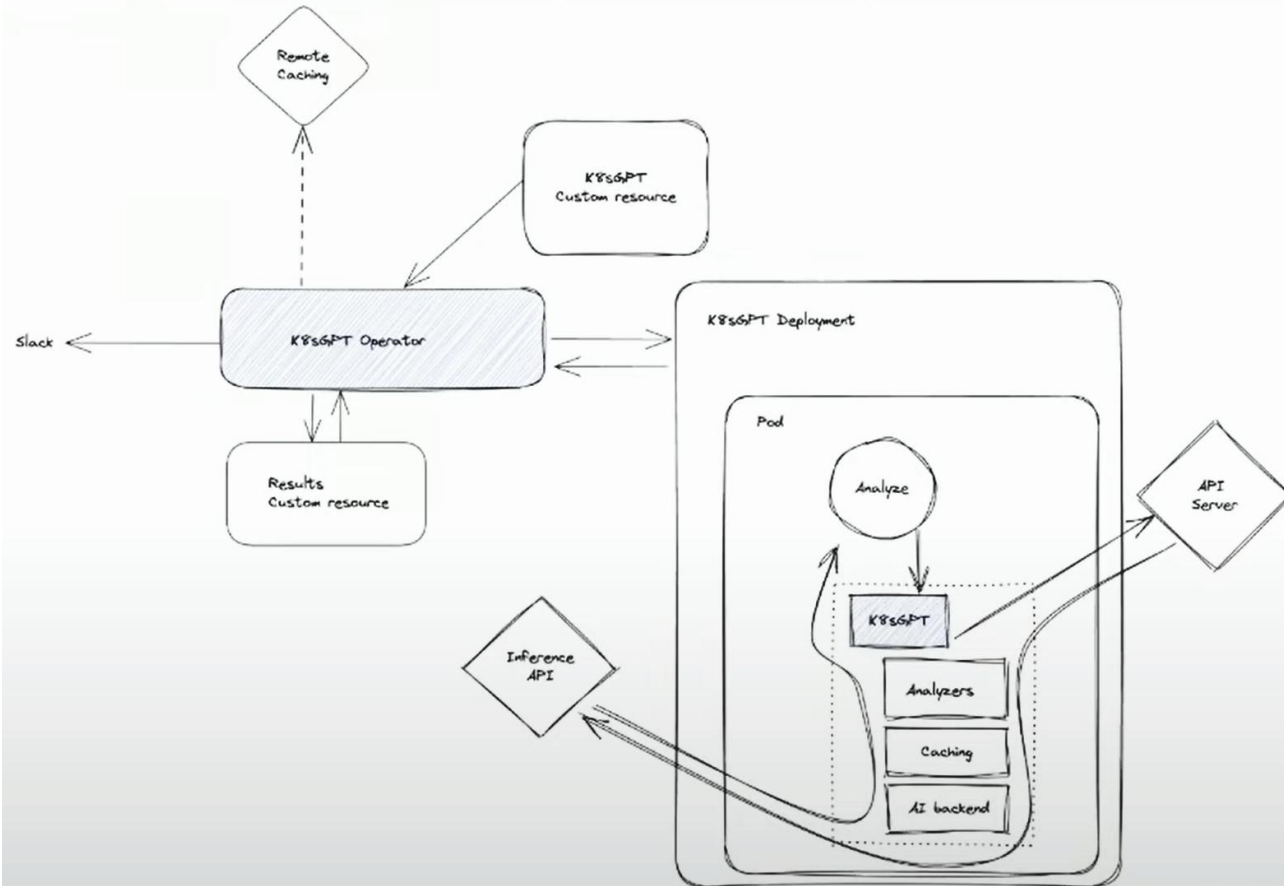
Name: defaultbrokenpod
Namespace: k8sgpt-operator-system
Labels: k8sgpts.k8sgpt.ai/backend=openai
        k8sgpts.k8sgpt.ai/name=k8sgpt-sample
        k8sgpts.k8sgpt.ai/namespace=k8sgpt-operator-system
Annotations: <none>
API Version: core.k8sgpt.ai/v1alpha1
Kind: Result
Metadata:
  Creation Timestamp: 2024-12-29T19:39:32Z
  Generation: 1
  Resource Version: 9414
  UID: 82512da5-567a-429a-b208-5273d82d94d0
Spec:
  Backend: openai
  Details: Error: The Kubernetes pod is unable to pull the specified image "nginx:1.a.b.c" due to back-off issue.

Solution:
1. Check the image name and version for typos.
2. Ensure the image "nginx:1.a.b.c" exists in the specified registry.
3. Verify the Kubernetes cluster's network connectivity.
4. Restart the pod or redeploy the application.

Error:
Text: Back-off pulling image "nginx:1.a.b.c"
Kind: Pod
Name: default/broken-pod
Parent Object:
Status:
Lifecycle: historical
Events: <none>
```



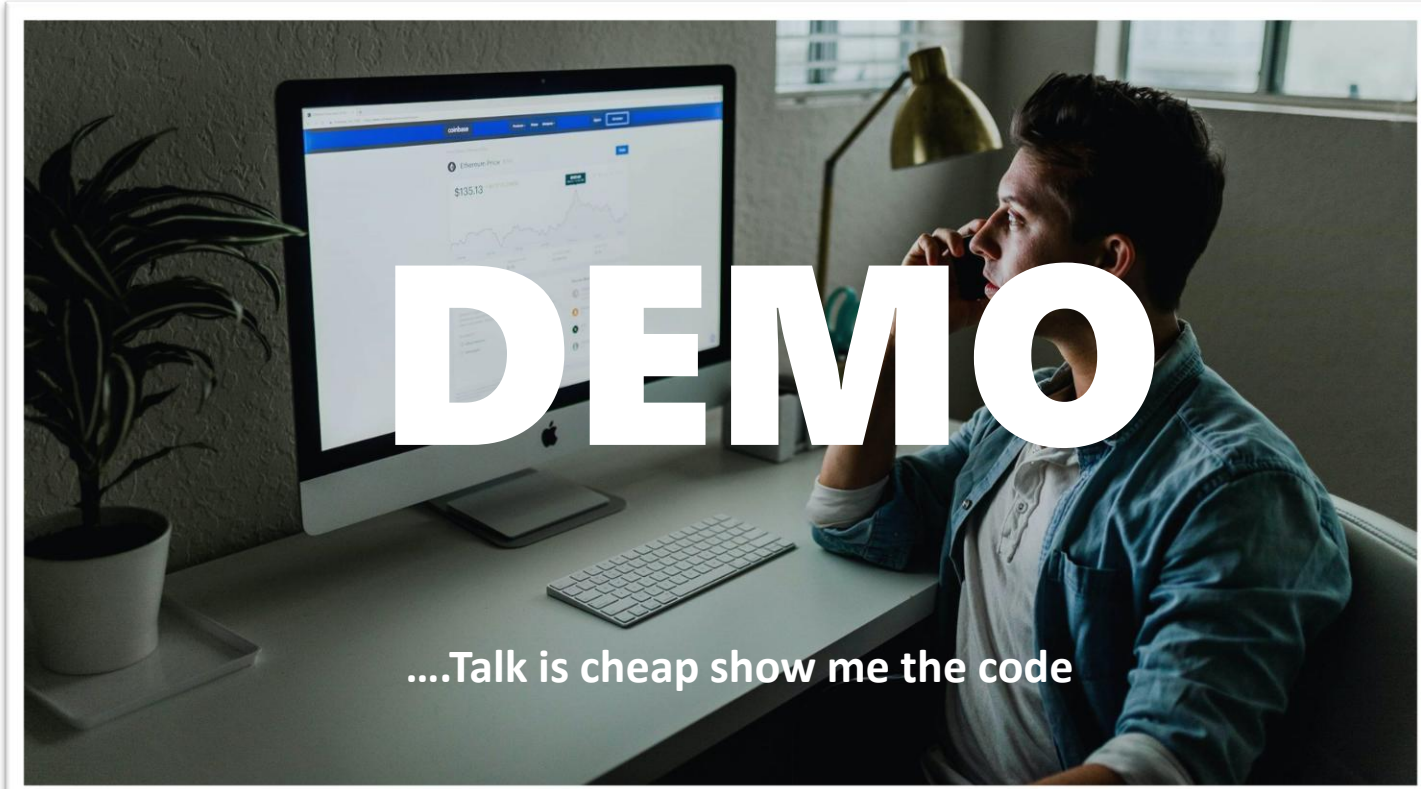
K8SGPT (Operator): Architecture



K8SGPT (Operator): Architecture

\$ h

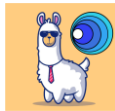
K8sGPT in practice



DEMO

....Talk is cheap show me the code

K8sGPT ...What's NEXT



NEW PROJECT



<https://github.com/schednex-ai/schednex>

THANK YOU



`Tech Beats unplugged`

Tech Beats Unplugged



Cloud Dude

Kosseila Hd

Principal Consultant @CloudThrill



Cloudthrill.ca



Twitter.com/CloudDude_



Linkedin.com/in/kousshd



GitHub.com/brokedba